

# CM0859 – MT5009 Type Theory Constructivism

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# Preliminaries

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## Textbook

“Lecture Notes on What is (Constructive) Logic?” (Pfenning 2023).

## Source code

The `HASKELL` code have been tested with GHC 9.12.2.

# Preliminaries

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## Logical constants

|             |                       |                         |
|-------------|-----------------------|-------------------------|
| $\wedge$    | (and)                 | conjunction             |
| $\vee$      | (or)                  | (inclusive) disjunction |
| $\supset$   | (if —, then —)        | implication             |
| $\perp$     | (falsity)             | bottom, falsum          |
| $\forall x$ | (for every $x$ )      | universal quantifier    |
| $\exists x$ | (there exists a $x$ ) | existential quantifier  |

# Preliminaries

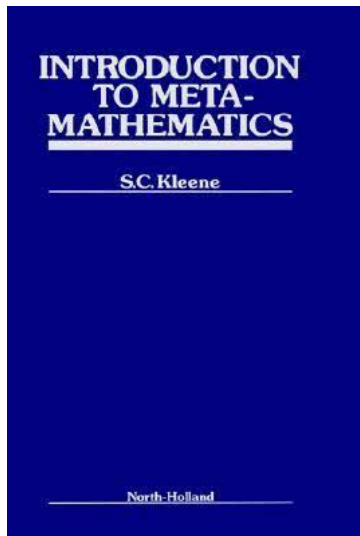
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## Definition

We define negation by  $\neg A := A \supset \perp$ .



# The Crisis in the Foundations of Mathematics

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Paradoxes  $\Rightarrow$  Crisis  $\Rightarrow$   $\left\{ \begin{array}{ll} \text{Logicism} & (\text{Russell and Whitehead}) \\ \text{Formalism} & (\text{Hilbert}) \\ \text{Intuitionism} & (\text{Brouwer}) \end{array} \right.$

# The Crisis in the Foundations of Mathematics

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## Logicism (Russell and Whitehead)

*“The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic.” (Kleene [1952] 1974, p. 43)*

# The Crisis in the Foundations of Mathematics

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## Formalism (Hilbert)

*“Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e. free from contradiction.”*

*(Kleene [1952] 1974, p. 53)*



# The Crisis in the Foundations of Mathematics

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## Intuitionism (Brouwer)

*“Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true.” (Iemhoff 2024)*

# The Crisis in the Foundations of Mathematics

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## Conceptions of the infinite

### (i) Non-Intuitionism

*“The infinite is treated as **actual** or **completed** or **extended** or **existential**. An infinite set is regarded as existing as a completed totality, prior to or independently of any human process of generation or construction, and as though it could be spread out completely for our inspection.”*  
(Kleene [1952] 1974, p. 48)

### (ii) Intuitionism

*“The infinite is treated only as **potential** or **becoming** or **constructive**. The recognition of this distinction, in the case of infinite magnitudes, goes back to Gauss, who in 1831 wrote, ‘I protest . . . against the use of an infinite magnitude as something completed, which is never permissible in mathematics.’ (Werke VIII p. 216.)”* (Kleene [1952] 1974, p. 48)

# Constructivism

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## Some differences with classical logic

- (i) Rejection of the principle of exclude middle (*tertium non datur*).

$$\vdash A \vee \neg A, \text{ for all formula } A.$$

# Constructivism

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- (i) Rejection of the principle of exclude middle (*tertium non datur*).

$$\vdash A \vee \neg A, \text{ for all formula } A.$$

- (ii) A proof of an existential formula  $\exists x.A(x)$  must include a **witness**  $t$  such as  $A(t)$  is true.

(continued on next slide)

# Constructivism

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## Some differences with classical logic

### (iii) Rejection of proofs by contradiction

Proof by contradiction  
(or *reductio ad absurdum*)

$$\frac{\begin{array}{c} [\neg A] \\ \vdots \\ \perp \end{array}}{A}$$

Proof of negation (Bauer 2017)

$$\frac{\begin{array}{c} [A] \\ \vdots \\ \perp \end{array}}{\neg A}$$

# Non-Constructive Proofs

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## Example

To prove that there are irrational numbers  $r, s \in \mathbb{R}$  such that  $r^s$  is rational.

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Proof (using the principle of exclude middle)

(whiteboard)

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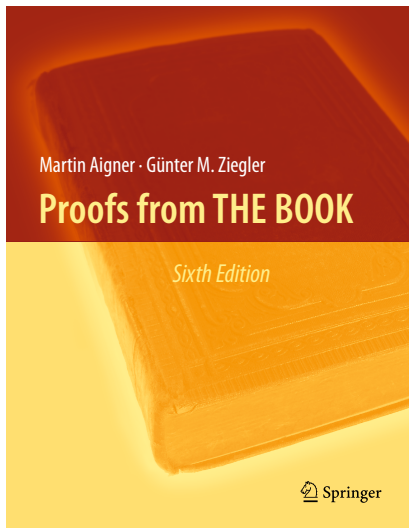
Proof (using the principle of exclude middle)

(whiteboard)

## Question

Could you give me two irrational numbers  $r, s$  such that  $r^s$  is rational?





# Non-Constructive Proofs

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To prove that there are an infinity number of primes.

# Non-Constructive Proofs

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## Proof (by contradiction)

**“Euclid’s proof.** For any finite set  $\{p_1, \dots, p_r\}$  of primes, consider the number  $n = p_1 p_2 \cdots p_r + 1$ . This  $n$  has a prime divisor  $p$ . But  $p$  is not one of the  $p_i$ : otherwise  $p$  would be a divisor of  $n$  and of the product  $p_1 p_2 \cdots p_r$ , and thus also of the difference  $n - p_1 p_2 \cdots p_r = 1$ , which is impossible. So a finite set  $\{p_1, \dots, p_r\}$  cannot be the collection of **all** prime numbers.”  
(Aigner and Ziegler [1998] 2018, p. 3)

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## Question

Could you give me an infinite list of primes?

# Non-Constructive Proofs

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## Observation

The axiom of choice is a source of non-constructive proofs.

# Non-Constructive Proofs

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## Definition

The **Cartesian product** (or **generalised product**) of a family of sets  $\langle A_i \mid i \in I \rangle$  is defined by

$$\prod_{i \in I} A_i := \left\{ f \mid f : I \rightarrow \bigcup_{i \in I} A_i \text{ and } \forall i (i \in I \rightarrow f(i) \in A_i) \right\}.$$

# Non-Constructive Proofs

## Definition

**Axiom of choice:** Let  $\langle H_i \mid i \in I \rangle$  be a family of sets. If  $H(i) \neq \emptyset$  for all  $i \in I$ , then  $\prod_{i \in I} H(i) \neq \emptyset$  (Enderton 1977).

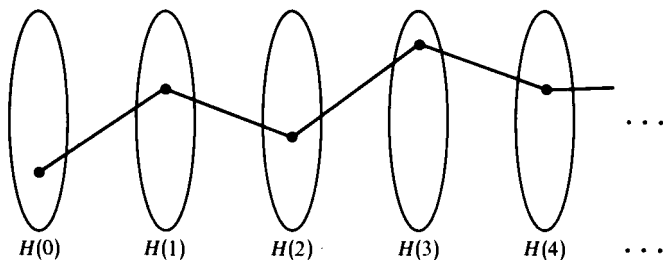


Illustration of the axiom of choice.<sup>†</sup>

<sup>†</sup>Figure source: (Enderton 1977, Fig. 11).

# The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

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## Constructive interpretation of the logical constants

- A proof of  $A \wedge B$  is a pair whose first component is a proof of  $A$  and whose second component is a proof of  $B$ .



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- A proof of  $\neg A$  is a function (method, program) which transforms a (hypothetical) proof of  $A$  into a contradiction.
- A proof of  $\exists x.A$  is a construction of a witness  $d$  and a proof of  $A(d)$ .
- A proof of  $\forall x.A$  is a function (method, program) which takes an arbitrary individual  $d$  into a proof of  $A(d)$ .

# Connection Between Proofs and Programs

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## Example

We define the follow predicates on natural numbers:

$$\text{even}(x) := \exists y. x = 2y,$$

$$\text{odd}(x) := \exists y. x = 2y + 1.$$

Prove that  $\forall x. \text{even}(x) \vee \text{odd}(x)$ .

Proof (by induction on  $x$ )

(whiteboard)

# Connection Between Proofs and Programs

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## Proof (by induction on $x$ )

(i) **Basis step:**  $x = 0$ .

Then  $\text{even}(x)$  is true because for  $y = 0$  (witness),  $x = 2y$ .

(ii) **Inductive step:**  $x = x' + 1$ .

For inductive hypothesis  $\text{even}(x') \vee \text{odd}(x')$  is true.

- **Case:**  $\text{even}(x')$  is true.

That is,  $x' = 2y'$  for some  $y'$ . Then  $x = 2y' + 1$  and therefore  $\text{odd}(x)$  is true and the witness is  $y'$ .

- **Case:**  $\text{odd}(x')$  is true.

That is  $x' = 2y' + 1$  for some  $y'$ . Then  $x = (2y' + 1) + 1 = 2(y' + 1)$  and therefore  $\text{even}(x)$  is true and the witness is  $y' + 1$ .



# Connection Between Proofs and Programs

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HASKELL function “from” the proof that  $\forall x.\text{even}(x) \vee \text{odd}(x)$

```
1  data Nat = Zero | Succ Nat
2
3  data E0 = Even | Odd
4    deriving Show
5
6  isEvenOrOdd :: Nat -> E0
7  isEvenOrOdd Zero      = Even
8  isEvenOrOdd (Succ n) = case isEvenOrOdd n of
9    Even -> Odd
10   Odd  -> Even
```

# Connection Between Proofs and Programs

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HASKELL function with witness “from” the proof that  $\forall x.\text{even}(x) \vee \text{odd}(x)$

```
1  data Nat = Zero | Succ Nat
2      deriving Show
3
4  data E0 = Even Nat | Odd Nat
5      deriving Show
6
7  isEvenOrOdd :: Nat -> E0
8  isEvenOrOdd Zero      = Even Zero
9  isEvenOrOdd (Succ x) = case isEvenOrOdd x of
10      Even y -> Odd y
11      Odd y  -> Even $ Succ y
```

## References

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